

Question ID: 8213b1b3

| Assessment | Test | Domain                            | Skill       | Difficulty                               |
|------------|------|-----------------------------------|-------------|------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Percentages | Hard <div></div> <div></div> <div></div> |

Question

According to a set of standards, a certain type of substance can contain a maximum of **0.001%** phosphorus by mass. If a sample of this substance has a mass of **140** grams, what is the maximum mass, in grams, of phosphorus the sample can contain to meet these standards?

Correct Answer: .0014

Rationale

The correct answer is **.0014**. It's given that a certain type of substance can contain a maximum of **0.001%** phosphorus by mass to meet a set of standards. If a sample of the substance has a mass of **140** grams, it follows that the maximum mass, in grams, of phosphorus the sample can contain to meet the standards is **0.001%** of **140**, or  $\frac{0.001}{100}(140)$ , which is equivalent to **(0.00001)(140)**, or **0.0014**. Note that .0014 and 0.001 are examples of ways to enter a correct answer.